ADESH KUMAR SAHOO

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SUMMARY

Applied Machine Learning and NLP practicioner focused on building production-oriented AI systems. Hands-on experience with retrieval-based NLP applications, predictive ML pipelines, LLM orchestration, evaluation frameworks, and modular backend deployment.

PROJECTS

LLM & RAG Evaluation Benchmarking Framework | GitHub | Demo

Sentence Transformers, OpenAI API, FastAPI, Streamlit

- Designed and deployed a benchmarking framework to systematically compare baseline LLM and RAG-based systems for QA evaluation.
- Built modular retrieval, generation, evaluation, and orchestration pipelines supporting both local and cloud inference workflows.
- Benchmarked using Exact Match, Semantic Similarity, Recall@k, MRR, and LLM-as-Judge evaluation.
- Improved semantic similarity by 58% and LLM-judged correctness by 35.8% with 95% Recall@k retrieval performance over baseline responses across a 150-sample benchmark.

Aggregate Summarizer (Pilot Service) | GitHub | Live Service

FastAPI, Redis, Docker, AWS EC2, OpenAI, Gemini

- Engineered and deployed a multi-LLM document summarization service on AWS EC2, orchestrating asynchronous summarization across multiple AI providers.
- Designed modular parsing, chunking, provider, evaluation, and consensus pipelines, using Redis for asynchronous job storage and automatic TTL cleanup.
- Reduced evaluation latency from ~3 minutes to under 10 seconds by profiling and optimizing the embedding pipeline on CPU-constrained infrastructure.

End-to-End Flight Delay Prediction System | GitHub | Demo

Scikit-learn, XGBoost, Streamlit, FastAPI

- Designed and deployed an end-to-end ML system for predicting flight delay risk, severity, and probable causes using pre-departure operational data.
- Built reusable preprocessing and inference pipelines by resolving training-serving skew, feature consistency, data leakage, and deployment constraints.
- Evaluated multiple ML models, demonstrating that feature engineering and not model complexity was the primary driver of predictive performance.

Context-Aware Business Japanese to English Interpreter Assistant (Ongoing Project)

- Currently building a manually curated bilingual business Japanese–English corpus (~500 sentence pairs) sourced from real business communication and annotated with metadata such as scenario, domain, and formality.
- Developing a context-aware business communication assistant using multilingual retrieval, metadata-aware ranking, and human-in-the-loop suggestion selection.

SKILLS

Languages: Python
Machine Learning: Feature Engineering, Model Evaluation, Pipelines
NLP and LLMs: Sentence Transformers, Semantic Retrieval, RAG, Embeddings, LLM Evaluation
Libraries: Scikit-learn, NumPy, Pandas, FastAPI
Deployment: Docker, AWS EC2, Render
Tools: Git, Redis, Jupyter

EDUCATION

Bachelor of Technology in Information Technology
Meghnad Saha Institute of Technology
2020 - 2024 | CGPA: 8.79

CERTIFICATIONS

Deep Learning Specialization - Andrew Ng (Coursera)
Machine Learning Specialization - Andrew Ng (Coursera)
Japanese Language Proficiency Test (JLPT) N2